

P vs NP: Research Synthesis and Project Audit

A critical review of the supplied research, local project notes, candidate constructions, and current literature

Prepared	23 September 2026	Research status	Open problem; no proof found
Inputs reviewed	Project audit dated 24 August 2026; earlier unusual-approaches research DOCX; locally mirrored files from the “P vs NP” and “P vs NP 2” ChatGPT projects; targeted primary literature and current public research indexes.		

Result in one sentence. This work found several exact, checkable lemmas and reproduced the project’s small-instance tests, but it did not produce a proof of $P = NP$ or $P \neq NP$. The strongest route remains a gap-MCSP lower bound through fusion/cover complexity; its missing step is still a new, nonlocal cover lower bound. The best near-term activity is to turn that bottleneck into a precise theorem with a representation-independent proof. A follow-up flow-support count also yields a conditional $O(k^5)$ bound on the rigid-pedigree set and polynomial size for the proposed input-dependent LPs; the membership and optimization proofs remain under audit.

1. Executive assessment

The Clay Mathematics Institute still lists P vs NP as unsolved. A claimed resolution would need a complete proof in a standard model, with all quantifiers and resource bounds explicit. No candidate in the materials reaches that bar.

The project is not empty of useful work. It has a strong organizing viewpoint: many approaches can be written as a search over fibers of a verifier, followed by a proposed certificate that a small representation cannot cover all hard instances. This unifies SAT objectives, circuit synthesis, compression, proof search, and cover methods. However, that viewpoint does not by itself make the witness findable, succinct, or cheaply verifiable.

The most rigorous positive items from this audit are:

- An exact expected-violation formulation for 3-SAT, together with a correct conditional-expectation rounding lemma. The global minimization remains NP-hard.
- A sound branch-collapse algorithm for XOR/NOT/AND circuits with running time $2^r \text{ poly}(N)$, where r is the affine dimension of the selected AND-control projection. It is polynomial only when $r = O(\log N)$.
- A simple full-span obstruction: truth tables of singleton inputs have circuits of $O(n)$ size and form a basis for the entire 2^n -dimensional truth-table space. This kills a direct linear-span/remote-point transfer to broad MCSP circuit classes.
- The project’s earlier code experiments pass their stated small-instance checks. These are validation of implementations, not evidence for an asymptotic separation.

The separator-to-filter shortcut remains circular: certifying high complexity and global semi-filter axioms is itself hard. Section 6 gives a constant-mass obstruction for canonical graph filters; Section 14 strengthens the canonical-only no-go to a single pair covering every canonical bit-literal filter in the Gap-MCSP magnification regime. Noncanonical semi-filters remain open.

New tests in Section 10 give project-level no-go lemmas for three specific semi-filter templates. Section 13 proves the partial-graph relaxation of promised Gap-MCSP has cover number at most one, and Section 14 gives a one-pair cover for the canonical bit-literal filters. These results rule out several restricted routes; the full promise-cover lower bound and P versus NP remain unresolved.

2. Scope, evidence, and limits

This synthesis combines the two files opened in the workspace with the locally available project mirrors. The mirrors contain research artifacts and scripts, including branch-collapse work and proof-deadline attempts; the available mirror did not contain a complete export of all private ChatGPT project conversations. That means this is a careful project audit, not a claim to have read every exchange or every paper ever written on P vs NP.

The literature pass targeted primary sources on hardness magnification, minimum circuit size, fusion/cover methods, remote points, constructive circuit refutation, proof complexity, proof-search deadlines, range avoidance, and known barriers. The latest ECCC reports and the Clay status page were checked through 23 September 2026. A literal survey of “all fields of mathematics and science” is neither possible nor useful here: P vs NP is a formal statement about classical computation, and a proposed cross-field connection must still imply a theorem in that model.

Labels used below: **proved** means a complete argument is supplied in this report or cited theorem; **tested** means finite experiments passed; **conditional** means the implication is valid only under a stated hypothesis; **conjectural** means the key statement remains unproved.

3. The shared fiber-and-avoidance viewpoint

Let $V(x,w)$ be a polynomial-time verifier, with instance x and certificate w . The accepting fiber is $F_x = \{w : V(x,w)=1\}$. NP asks whether F_x is nonempty. Equivalent representations include satisfying assignments, paths through a computation, low-complexity descriptions, proof strings, and preimages of circuit outputs.

Many project ideas add a second object: a compact representation, a distribution, a cover, a proof-search clock, or a structural invariant that is supposed to expose whether F_x is empty. The recurring obligation is to show all three parts:

1. the proposed object exists for every input in the intended case;
2. it can be constructed or recognized within the claimed resource bound;
3. its guarantees do not already assume the hard answer (for example, that a truth table has large circuit complexity).

This is a useful diagnostic framework, but not a new separation theorem by itself. It explains why several attractive ideas collapse back to SAT, MCSP, or global proof verification.

4. Exact constructions and their limits

4.1 Expected violations for 3-SAT

For a 3-CNF formula F and independent Boolean variables with probabilities $p = (p_1, \dots, p_n)$, define $E_F(p)$ as the expected number of violated clauses. Each clause contributes a product of at most three terms, each term being p_i or $(1-p_i)$, so E is a degree-at-most-three multilinear polynomial.

F satisfiable $\Leftrightarrow \min_{p \in [0,1]^n} E_F(p) = 0$. If F is unsatisfiable, $E_F(p) \geq 1$ for every p .

If a supplied p has $E_F(p) < 1$, sequential conditional-expectation rounding fixes variables without increasing the expected number of violations. At the end, the number of violated clauses is an integer below one,

hence zero. This is a rigorous witness extraction lemma. Finding a global minimizer or any p with value below one remains NP-hard because integral p already encodes SAT.

4.2 Circuit fibers and tensor networks

A Boolean circuit can be encoded by one local relation per gate. Keeping internal wire values gives an exact $O(s)$ -size constraint/tensor representation for a circuit with s gates. The zero test for the resulting network is SAT-complete; exact contraction is #SAT/#P-hard. The representation is faithful, but the apparent local structure does not yield a general polynomial-time contraction method.

4.3 Affine control rank for XOR/NOT/AND circuits

After linear wiring constraints are collected, choose one Boolean control operand for each AND gate. Fixing all controls makes every AND relation linear in the remaining wire variables. If the feasible control vectors occupy an affine subspace of dimension r , enumerating its 2^r points and solving the resulting linear systems gives a sound $2^r \text{ poly}(N)$ -time procedure for the represented satisfiability task. It is polynomial for logarithmic r . There is no bound in the project that forces $r = O(\log N)$ for arbitrary circuits, so the worst case remains exponential.

5. Tested results from the project

Check	Observed result	What it establishes
3-variable 3-SAT energy and rounding	All 256 subsets of the 8 possible width-3 clauses were enumerated; 6,912 probability-grid cases were checked. For all 5,023 tested points with $E < 1$, rounding found a satisfying assignment.	Confirms the finite implementation checks and the stated small-instance lemma. The general lemma follows from conditional expectation, not from the experiment.
Affine collapse solver	180 random circuits (146 satisfiable, 34 unsatisfiable) and 120 random affine/AND systems agreed with exhaustive search in every case.	Evidence against implementation bugs in sampled small cases only.
Branch-collapse experiments	9 parity cases and 16 equality-ordering cases passed. Unary propagation stalled on 3 inconsistent parity instances; Gaussian elimination found the contradictions. A blocked ordering retained 2,4,...,256 residuals for $m=8$, while an interleaved ordering stayed at at most 2.	Shows that linear algebra can remove parity structure missed by unary propagation, and that variable order can dominate residual width. No general width bound follows.
Linear-span / remote-point transfer	For $n=1,...,5$, singleton truth tables had full $GF(2)$ rank; the general proof is immediate from their indicator circuits.	Rules out the naive claim that all small-circuit truth tables lie in a proper linear subspace once the size threshold includes $O(n)$ -gate singleton indicators.

The project scripts were copied to a temporary directory before rerunning. The user's project files were not modified. Small random or exhaustive tests can detect errors; they cannot establish behavior for all input lengths.

6. The strongest conditional route: Gap-MCSP and fusion covers

Hardness magnification gives an unusually short bridge from a weak lower bound on a meta-computational problem to a major circuit separation. Oliveira, Pich, and Santhanam prove that for a universal constant c , if there is $\epsilon > 0$ such that for every sufficiently small $\beta > 0$, the gap problem $\text{Gap-MCSP}[2^{\beta n}/(cn), 2^{\beta n}]$ is not in $\text{Circuit}[N^{1+\epsilon}]$ (where $N=2^n$), then NP is not contained in polynomial-size circuits, and therefore $P \neq \text{NP}$.

The project proposed proving the weak lower bound through a two-dimensional cover problem. Cavalier and Oliveira formalize the fusion/cover framework and show how graph covers transfer to circuit lower bounds. The cofactor test in Section 13 shows that the unrestricted row/column proxy for this Gap-MCSP promise has cover number at most one. Only the direct bit-literal promise-cover lower bound remains a live version of this route; it is still unproved.

A fractional-cover dual remains a possible method: a distribution over noncanonical semi-filters that gives every pair bad mass at most $N^{-(1-\epsilon)}$ would force the required lower bound. The canonical subclasses are ruled out by one-pair covers in Section 14, and the unrestricted graph proxy is trivial by Section 13. No suitable noncanonical distribution or uniform pair bound is known.

No-go for canonical-filter distributions

Let G be a bipartite graph on $[N] \times [N]$, let U be its nonedges, and assume every row and column contains at least $d \geq 1$ points of U . For each edge $e=(u,v)$ in G , write R_u and C_v for its row and column sets inside U . The canonical semi-filter above e is $F_e = \{W \subseteq U : R_u \subseteq W \text{ or } C_v \subseteq W\}$. Take any probability distribution μ on edges of G and view it as a distribution on the corresponding canonical filters (merging duplicates).

Choose independent uniformly random subsets S of rows and T of columns, and set $E = \bigcup_{i \in S} R_i$ and $H = \bigcup_{j \in T} C_j$. For a fixed edge $e=(u,v)$, condition on $u \in S$ and $v \in T$; then $E, H \in F_e$. The pair covers F_e exactly when $E \cap H \not\subseteq F_e$. This requires that not all nonedges in row u have their columns in T , and not all nonedges in column v have their rows in S . The two containment events (all row nonedges in T and all column nonedges in S) are independent, with probabilities $2^{-(|R_u|)}$ and $2^{-(|C_v|)}$.

$$\Pr[(E,H) \text{ fails to preserve } F_e] = 1/4 \cdot (1 - 2^{-(|R_u|)}) \cdot (1 - 2^{-(|C_v|)}) \geq 1/4 \cdot (1 - 2^{-(d)})^2.$$

Averaging this bound over μ shows that some pair (E,H) is covered by at least $1/4 \cdot (1 - 2^{-(d)})^2$ of the μ -mass. For $d \geq 2$ this is at least $9/64$, so μ cannot certify a cover lower bound growing with N . The same argument works for arbitrary weights μ , not just the uniform distribution.

Finite check: for the 4×4 diagonal graph, its complement has minimum row/column degree 3. I enumerated all 256 choices of S, T and 1,000 seeded random distributions on the four canonical filters; the best pair covered all four filters in every trial. This checks the code on one example; the general claim follows from the proof above.

Application to the selected Gap-MCSP encoding. Let $N=2^n$ be the full truth-table length and $K=2^{(N/2)}$ the number of half-tables on each side. For $s_2=N^\beta$ with fixed $\beta < 1$, the number C of n -input circuits of size below s_2 is at most $2^{(O(s_2 \cdot \log(n+s_2)))} = 2^{o(N)}$. Each fixed half-table has K completions, of which at most C are below s_2 . Thus every row and column has at least $d=K-C=K(1-o(1))$ high-side completions. The lemma gives some fusion pair covering at least $1/4 \cdot (1 - 2^{-(d)})^2 = 1/4 - o(1)$ of any measure supported only on canonical filters. This rules out the canonical-only fractional-dual strategy for the target Gap-MCSP graph at its magnification parameters.

This obstruction applies only to measures supported on canonical filters; distributions that put substantial mass on noncanonical semi-filters remain open. It is a project-level lemma and no claim of novelty in the literature is made.

Needed statement	Status
Correct parameter transfer from the exact cover problem to the selected Gap-MCSP family	Known frameworks exist; each parameter and representation must be matched precisely.
$\rho > N^{1+\epsilon}$ for explicit instances uniformly over the small- β regime	Open target; no proof found.
Apply the magnification theorem	Conditional theorem is known once the preceding lower bound has exactly the required form.

7. Why “separator implies filter” is not yet a construction

A proposed route starts with a small circuit D purporting to separate low-complexity from high-complexity truth tables, then tries to construct a semi-filter F that defeats D on a fusion pair. If D really solves the promise problem, a suitable contradiction would refute it. But the procedure must work without knowing whether a truth table is in the high-complexity promise region.

There are two concrete circularity points:

- **Hard-side certification.** Membership in the low-complexity side has a short circuit witness. A general truth table’s circuit complexity being at least s_2 has no known short certificate suitable for this construction.
- **Succinct global validity.** An explicit semi-filter over all subsets of a large ground set is enormous. A succinct representation must still guarantee exclusion of the empty set, the threshold condition, upward closure, and pair-preservation over an enormous family of subsets. Checking a few sampled sets does not establish these properties.

Therefore the “separator-to-filter” idea is a useful theorem schema, but the missing construction is at least as hard as the lower bound it is intended to prove. A tangible improvement would be a compact certificate format with a sound local verification theorem whose proof avoids testing all subsets.

8. Other project routes and external barriers

Proof-deadline / short-proof route

The project defines a sentence D_n asserting that a fixed proof-searcher does not find a proof by time 2^n . Under consistency, each fixed D_n can be proved using a finite nonhalting trace, but that does not control the proof length s_n . The project’s reduction establishes a conditional route of the form: consistency of the base theory plus sufficiently small proof lengths infinitely often implies $P \neq NP$. The needed short-proof infinitely-often hypothesis is unproved. A fast uniform proof generator is constrained by the deadline argument, but this leaves open slow or nonuniform short proofs. Oracle variants show that deadline/proof-length behavior can change with the formal system; they are not a refutation of the PA-specific statement.

Range avoidance and constructive refuters

Range Avoidance asks for an object outside the range of a succinct map and connects to one-way functions, proof systems, and lower bounds. The project's single-valued and demi-bit variants are worth tracking, but no route in the supplied notes converts their current guarantees into a P-vs-NP separation. Recent gate-elimination work demonstrates constructive refuters for XOR, multiplexers, and affine dispersers. Those results sharpen the lesson: a lower-bound proof can become an algorithm only when its elimination steps expose a violation efficiently. The paper's optimal XOR case depends on a full classification of optimal circuits, a feature not available for SAT or general functions.

Classical barriers

Relativization, natural-proofs, and algebrization results rule out broad families of proof strategies under their respective hypotheses. They do not prove that $P \neq NP$ is impossible. Any proposed argument should be checked for oracle relativization, largeness/usefulness in the natural-proofs sense, and algebraic relativization. The project's broad topology, sheaf, semigroup, quantum, statistical-physics, and nonstandard-model ideas do not currently bypass these checks; they need a precise lemma in the standard deterministic Turing-machine model first.

9. New obstruction from the linear-span test

For every $a \in \{0,1\}^n$, let δ_a be the truth table that is 1 at a and 0 elsewhere. It can be computed by ANDing the n literals that specify $x=a$, using $n-1$ AND gates (plus at most n input negations). The 2^n functions δ_a are the standard basis vectors of the truth-table vector space over every field. Hence:

For any field K and any circuit-size bound $s \geq n-1$ plus literal-negation cost, the K -linear span of size- s truth tables is the entire K^{2^n} .

This is a complete obstruction to a direct method that tries to find a remote truth table outside the linear span of all low-complexity truth tables at such thresholds. It does not rule out remote-point methods using nonlinear structure, restricted subfamilies, or a different feature map. It is recorded as a project-level lemma, not claimed as a new result in the literature.

10. Research directions ranked by proof leverage

1. **Top priority: make the fusion target exact.** Write down the ground set, low/high promises, cover vertices, semi-filter axioms, and reductions with all constants and quantifiers. Then identify the weakest lemma that would imply the required $N^{1+\epsilon}$ bound. This prevents hidden changes of problem between the cover theorem and hardness magnification.
2. **Search for a genuinely local dual witness.** Try distributions or potentials over semi-filters whose pairwise bad-mass bound follows from a checkable invariant of the explicit truth table. The invariant must be sound over all pairs, not sampled pairs. Stress-test it first on structured examples and on oracle-gate/adaptive constructions.
3. **Extend branch collapse only where rank is structurally bounded.** Investigate circuit families with limited AND-control rank, low treewidth after affine elimination, or restricted fanout. A useful theorem must state the structural promise and prove the bound on every input in the family.
4. **Turn proof deadlines into explicit proof-complexity bounds.** Isolate the formal-theory assumptions and seek a finite-consistency consequence strong enough to constrain the exact proof-length sequence. The current gap is not proof-search runtime but the missing theorem about short proofs.
5. **Use constructive lower bounds as a model.** For any proposed combinatorial lower bound, extract an algorithm that refutes a candidate small circuit. If the "refuter" needs an uncomputable or globally hard choice, that is the bottleneck to solve.

Stop a branch when its next lemma is equivalent to SAT, MCSP, a general circuit lower bound, or global verification over all subsets, unless a new invariant makes that lemma easier. This is not a claim that those avenues are impossible; it makes the actual missing work explicit.

Rectangle-closure filters fail under rowwise splitting

Consider the proposed family F_e^{rect} of all sets W contained in U that contain the nonempty hole set $U \cap Q$ for some combinatorial rectangle Q containing e . This family is upward closed and excludes the empty set. In the project graph, where every row and column has many holes, it also contains the row-star and column-star required for a semi-filter above e .

Assume every row of U has at least two holes. Split the holes in each row into two nonempty parts, and let E and H be the unions of the first and second parts. Then E and H are disjoint. For any positive edge $e=(u,v)$, select a hole (u,a) from the first part of row u . The rectangle consisting of cells (u,v) and (u,a) has exactly the nonempty hole set $\{(u,a)\}$, so E belongs to F_e^{rect} . A hole from the second part gives the same argument for H . But $E \cap H$ is empty, which belongs to none of these filters. Thus this one pair violates preservation for every positive edge simultaneously.

Consequently, every probability distribution over these rectangle-closure filters has a pair covering mass 1; it cannot yield a useful fractional-dual lower bound. In the Gap-MCSP encoding from Section 6, circuit counting gives at least $K-C=K(1-o(1))$ high-complexity completions in each row, so the row-degree condition holds by a wide margin. As a finite sanity check, exact enumeration of all pairs E,H on 24 sampled 4-by-4 graphs (each with 5 to 8 positive edges and complement row and column degrees at least 2) also found full coverage; the argument above explains this outcome generally. This rejects this particular noncanonical construction, not the Gap-MCSP route or P versus NP.

Threshold filters have a constant-mass obstruction

For each edge $e=(u,v)$, let $d_r(u)$ and $d_c(v)$ be the numbers of holes in its row and column. Choose thresholds $1 \leq a_u \leq d_r(u)$ and $1 \leq b_v \leq d_c(v)$, and define $F_e = \{W \subseteq U : |W \cap R_u| \geq a_u \text{ or } |W \cap C_v| \geq b_v\}$. This is an upward semi-filter above e because it contains both stars and excludes the empty set.

Here is a constant-mass bound for this entire threshold family. Fix any probability distribution μ on the positive edges. Let L_r be the mass of edges whose row threshold is at most half its row degree, and L_c the mass whose column threshold is at most half its column degree. If either mass is at least $1/3$, partition the holes so that each such row, or column, is split into two parts each meeting its threshold. The resulting disjoint pair E,H belongs to every corresponding filter, while $E \cap H$ is empty, so it covers at least $1/3$ of μ . Otherwise, more than $1/3$ of μ lies on edges for which both thresholds exceed half their degrees. Choose a random set S of rows and a random set T of columns, independently including each vertex with probability $1/2$. Let E be the union of the hole-stars of rows in S , and H the union of the hole-stars of columns in T . For a high-threshold edge $e=(u,v)$, conditional on u in S and v in T , E and H belong to F_e . In $E \cap H$, the row and column star counts are independent binomial random variables with parameters $(d_r(u), 1/2)$ and $(d_c(v), 1/2)$. Each is below its threshold with probability at least $1/2$. Including the probability $1/4$ that u in S and v in T , e is covered with probability at least $1/16$. The expected covered mass therefore exceeds $1/48$, so some pair covers at least $1/48$ of μ .

Thus no choice of per-row and per-column thresholds in this family can provide an inverse-polynomially small maximum covered mass. An exact finite check enumerated all pairs E,H on 30 seeded 4-by-4 graphs (8 to 10 holes, minimum complementary row and column degree 2), using random thresholds and random edge weights; the best pair covered between 0.890

and 1.000 of the mass in every instance. This no-go applies to threshold-only filters, not arbitrary semi-filters. It is an internal project lemma; no claim of literature novelty is made.

Cross-support filters also fail under a two-color split

Define $F_e^{\text{cross}} = \{W \subseteq U : R_u \text{ is a subset of } W \text{ or } C_v \text{ is a subset of } W \text{ or both } W \cap R_u \text{ and } W \cap C_v \text{ are nonempty}\}$, for $e=(u,v)$. In the dense-complement graph this is an upward semi-filter above e and excludes the empty set. It adds a natural cross-coordinate condition to the mandatory row-star and column-star sets.

Color every hole independently red or blue. If every row and column has at least d holes, the probability that any of the $2K$ rows or columns is monochromatic is at most $2K$ times $2^{-(1-d)}$. When this quantity is below 1, a coloring exists in which every row and column contains both colors. Let E and H be the red and blue holes. For every positive edge $e=(u,v)$, both E and H hit its row-star and its column-star, so both belong to F_e^{cross} . Since $E \cap H$ is empty and the filters exclude the empty set, the same pair covers all positive edges.

For the Gap-MCSP graph, Section 6 gives $d \geq K-C = K(1-o(1))$; hence $2K$ times $2^{-(1-d)}$ is below 1 for sufficiently large instances. A finite exact check also found a valid split in each of 30 seeded 4-by-4 graphs (seed 674912; 8 to 10 holes and minimum row and column degree 2), with the maximum covered mass equal to 1 in every graph. This rejects this cross-support family in the target regime, while leaving arbitrary semi-filters open.

11. What would count as a real breakthrough

For $P \neq \text{NP}$: a complete, uniform lower bound for an NP-complete problem against all polynomial-time algorithms, or a theorem implying NP is not in P/poly with a fully checked reduction. For $P = \text{NP}$: a deterministic polynomial-time algorithm for SAT (or another NP-complete language) with a correctness proof and a polynomial running-time bound. A finite experiment, heuristic, average-case result, restricted circuit lower bound, or conditional theorem is useful but does not settle the question.

At the intermediate level, a new theorem is substantial if it proves an explicit superlinear cover lower bound in the exact magnification regime, constructs a succinct sound semi-filter, proves a structural rank bound for a meaningful general circuit class, or derives a genuinely stronger proof-length consequence. None of those has been obtained here.

12. Selected primary references

- [Clay Mathematics Institute, P vs NP](#) — current official problem status and statement.
- [Oliveira, Pich, Santhanam, “Hardness Magnification Near State-of-the-Art Lower Bounds” \(Theory of Computing, 2021\)](#) — exact Gap-MCSP parameters and magnification implication.
- [Cavalar and Oliveira, “Boolean Circuit Complexity and Two-Dimensional Cover Problems” \(2025\)](#) — fusion, cover complexity, cyclic characterization, and transfer to circuit lower bounds.
- [Volk, “Improved Algorithms for the Remote Point Problem” \(ECCC TR26-203, 22 September 2026\)](#) — current remote-point result relevant to the project’s linear-span idea.
- [Carmosino, Dang, Jackman, “Constructive Separations from Gate Elimination” \(2026 preprint\)](#) — algorithms extracting counterexamples from several circuit lower-bound arguments.
- [McKay, Murray, Williams, “Weak Lower Bounds on Resource-Bounded Compression Imply Strong Separations of Complexity Classes” \(STOC 2019\)](#) — search-MCSP/streaming magnification.
- [Chen, Jin, Santhanam, Williams, “Constructive Separations and Their Consequences” \(2022\)](#) — constructive lower bounds and their implications for class separations.
- [Li, “Symmetric Exponential Time Requires Near-Maximum Circuit Size: Simplified, Truly Uniform” \(2023; revised 2024\)](#) — single-valued Range Avoidance and near-maximum circuit lower bounds.
- [Ren, Wang, and Zhong, “Hardness of Range Avoidance and Proof Complexity Generators from Demi-Bits” \(ITCS 2026\)](#) — conditional range-avoidance hardness and proof-generator results.

• [Krajíček, “Information in Propositional Proofs and Algorithmic Proof Search” \(2021\)](#) — proof information and bounded proof-search context.

• [Pich, “Localizability of the Approximation Method”](#) (Computational Complexity 33, 2024) — locality limits for approximation-method lower bounds with magnification.

• [Aaronson and Wigderson, “Algebrization: A New Barrier in Complexity Theory” \(2008\)](#); [Razborov and Rudich, “Natural Proofs” \(1997\)](#); [Baker, Gill, and Solovay, relativization \(1975\)](#) — standard barrier checks.

13. Structural correction: the graph relaxation collapses

Status: proved obstruction to one proof route; this is not a P versus NP result.

Proposition. Let $N = 2^n$ be the truth-table length, and split an n -variable truth table into cofactors (x,y) on $n-1$ variables. Let Y contain the tables of circuit size at most s_1 , let Z contain those of size at least s_2 , and let $\Gamma = Y \cup Z$. If $2s_1 + O(1) < s_2$, and L and R are the left and right projections of Y , then $Y = (L \times R) \cap \Gamma$.

Proof. Restricting a size- s_1 circuit to either value of its first input computes each cofactor with at most $s_1 + O(1)$ gates. Conversely, for any $x \in L$ and $y \in R$, a selector on the first input combines circuits for x and y using at most $2s_1 + O(1)$ gates, below s_2 . Their concatenation is not in Z ; if it belongs to Γ , it must be in Y .

Consequence. With an unrestricted row star and column star for every graph vertex as generators, $Y = (\text{union of rows indexed by } L) \text{ intersected with } (\text{union of columns indexed by } R)$, on Γ . Thus graph intersection complexity is at most one, and graph fusion-cover complexity is at most one. This rules out the proposed $N^{(1+\epsilon)}$ lower bound for that graph relaxation.

Scope. This does not give a small circuit over the original $2n$ bit-literal generators: arbitrary sets L and R are free unions only in the graph model. The direct bit-literal promise-cover lower bound remains unresolved.

This cofactor obstruction is an internal result of the audit, with no claim of literature novelty. The fusion definitions follow Cavalar and Oliveira (2025); the parameter gap follows Oliveira, Pich, and Santhanam (2021).

14. Exact one-pair cover of canonical bit-literal filters

Let U be the high-complexity side of Gap-MCSP and, for each low-complexity truth table a , let F_a^{can} consist of all sets containing at least one literal slice $B_{(i,a_i)}$ intersect U . This is the minimal semi-filter above a . For sufficiently large N , every one-coordinate high-complexity slice is nonempty by circuit counting.

Circuit counting gives at most $2^{O(s_1 \log(n+s_1))}$ low-complexity truth tables. Choose disjoint sets I, J of t truth-table positions, with $t = O(s_1 \log(n+s_1))$, and patterns p on I and q on J that no low-complexity truth table realizes.

Let E contain high truth tables that disagree with p on I , and let H contain high truth tables that disagree with q on J . Every low table disagrees with both patterns, so E and H each contain one of its required literal slices. For any coordinate k , a high table can be chosen to match the low table at k and match the forbidden pattern on the other block. Counting shows such a high completion exists because $2^{(N-t-1)}$ exceeds the number of circuits below s_2 . Thus no required literal slice is contained in $E \text{ intersect } H$.

Therefore one pair (E,H) fails preservation for every canonical bit-literal filter. This improves the earlier constant-mass bound for canonical-only distributions. The intersection $E \text{ intersect } H$ is nonempty, and arbitrary noncanonical semi-filters can include it, so the result does not

bound full fusion-cover complexity. This is an internal project lemma with no claim of literature novelty and no implication $P \neq NP$.

15. Pedigree flow bound and extension-complexity audit

Research follow-up, 23 September 2026. This update strengthens the size analysis of the 2025 pedigree-polytope membership proposal and checks a proposed extension-complexity objection. It is a conditional algorithmic-size result, not a proof of $P = NP$ or $P \neq NP$.

An independent bound on rigid pedigrees

Let r_k be the number of rigid pedigrees R_k . In F_k , split the origins into residual edge origins U_k and pedigree origins $R_{(k-1)}$, and let D_k be the destinations. For a basic feasible transportation flow, the arcs strictly between their lower and upper bounds form a forest: an alternating perturbation around an interior cycle would preserve all balances and contradict extremality.

A pedigree origin P has supply μ_P and each outgoing arc has capacity μ_P . Thus at most one positive arc leaving P can be at its upper bound. If P has positive interior arcs, it cannot also have a positive upper-bound arc; it must have at least two interior arcs, because a sole interior arc would carry all μ_P and be at its upper bound. In each component of the interior-arc forest, with p pedigree origins and d destination nodes, every pedigree origin has degree at least two and the forest has at most $p+d-1$ edges. Therefore $p \leq d-1$ and there are at most $2d-2$ such edges in the component. Summing over components, at most $2|D_k|$ positive interior arcs leave pedigree origins.

Each positive rigid arc from a pedigree origin contributes at most one extension to R_k . Each possible residual-origin/destination pair contributes at most one pedigree, so $r_k \leq r_{(k-1)} + 2|D_k| + |U_k||D_k|$. Since $|U_k| \leq \text{binom}(k-1, 2)$, $|D_k| \leq \text{binom}(k, 2)$, and $r_4 \leq 18$, iteration gives $r_k = O(k^5)$. This argument replaces the paper's stated clique-to-simplex and weight-fiber dimension inference, conditional on its flow construction and one-pedigree-per-rigid-arc rules.

The size of the layered networks is then polynomial: $O(n^6)$ pedigree nodes in total and $O(n^8)$ arcs across layers. There are $O(n^4)$ restricted-network max-flow calls per stage. The multicommodity formulation uses at most one commodity per network arc; even counting one variable for every commodity-arc pair gives $O(n^{16})$ variables and polynomially many constraints. These bounds repair the polynomial-size premise only. The membership characterization and its conversion to STSP optimization still need complete verification.

Why extension complexity is not a direct refutation

An extended formulation is one fixed polyhedron with one fixed linear projection onto the target polytope. The pedigree procedure instead builds the network, rigid sets, capacities, and multicommodity LP from the queried point X and earlier X -dependent flows. A point-dependent LP feasibility test does not automatically give a fixed extended formulation for the whole polytope.

Rothvoss proved exponential extension complexity for the perfect-matching polytope and, by a known reduction, for the TSP polytope. Perfect matching is nevertheless polynomial-time solvable, illustrating that efficient optimization and small extended formulations are different properties. See [Rothvoss's extension-complexity paper](#) and [Edmonds's maximum-matching paper](#). Thus, the claim that polynomial-time membership implies a compact extended formulation is not valid as stated. A contradiction would require one fixed polynomial-size

LP whose projection is the entire pedigree polytope; the paper does not present such an LP. Rejecting that objection does not establish the membership algorithm's correctness.

Correction and remaining obligations

The 2025 preprint explicitly restricts to commodities with positive flow before introducing normalized vectors, so the zero-flow division issue raised in the earlier audit is withdrawn. The formal development should preserve that restriction. The main open proof obligations remain the exact necessity and sufficiency of the multicommodity-flow membership test, the polynomial bit-complexity of every construction on rational input, and the membership-to-separation and STSP optimization implications.

Primary source for the membership and complexity construction: [Arthanari, On the Importance of Studying the Membership Problem for Pedigree Polytopes \(2025\)](#). The updated project audit in `P_VS_NP_PROJECT_AUDIT_2026-08-24.md` contains the full derivation and qualifications.

17. Audit of the public Lean formalization

Audit date: 23 September 2026. The public Lean repository does not currently provide a machine-checked proof of $P = NP$. Its main theorem is derivable only after assuming a custom axiom that directly asserts polynomial-time pedigree-polytope membership. The Lean MCF data structure also imposes a one-hot restriction on the target layer, unlike the general fractional-demand MCF in the paper. This is a concrete defect in the formalization and its claim of verification; it is not evidence for either $P = NP$ or $P \neq NP$.

The decisive complexity premise is an axiom

In `N_PEqualsNP.lean`, `QuickProtocol` is itself declared as an axiom, without a definition connecting it to an executable algorithm. The further axiom `tardos_strongly_polynomial` has the conclusion `QuickProtocol (LayeredPoint n)`. The main theorem then takes that proposition as an input to the Maurras and GLS steps, followed by the reduction from pedigree optimization to STSP and the standard NP-completeness implications. Therefore the formal proof assumes the key polynomial-time membership claim instead of deriving it from the proposed MCF construction. Citing Tardos does not discharge this obligation: the formal development must state the actual LP and prove that its feasibility decides membership in both directions, with polynomially bounded construction and bit complexity.

The repository's own README states that the necessity direction of the MCF characterization is an axiom and that a backup proof contains 16 sorry placeholders. Although a sufficiency direction could suffice for sound positive answers, it cannot by itself justify a membership decision procedure: completeness / rejection correctness must also be established. The separate axiom above bypasses that missing direction in the $P=NP$ chain.

The formal MCF forces a one-hot target layer

`N_LayeredNetworkTypes.lean` defines `MCFFeasible.src_val` as $\sum s.\text{arc_head} = 1$ for every commodity. The same structure requires all coordinates of the target layer to be nonnegative and to sum to 1. It follows that every commodity head must be the same single triple: two distinct heads with value 1 would already contribute 2 to a layer sum of 1. Thus this structure cannot represent an input whose final layer has two or more positive coordinates.

In the paper, the final-layer demand is distributed among destinations, and each commodity's positive flow is a fractional quantity. The paper's later normalized vector $\mathbf{Y}^s / \mathbf{v}_s$ has a unit vector at the final layer; that is a property of the commodity-normalized vector, not of the

original input X . The Lean field places this unit-value condition on X itself. `N_YisinConv.lean` then calls the packability corollary on X and uses this condition to obtain a special one-destination case.

A member that the Lean MCF cannot represent

The 2025 preprint Example 8 has $x_4 = (0, 3/4, 1/4)$, $x_5 = (1/2, 0, 0, 0, 0, 1/2)$, and $x_6 = (1/8, 1/8, 3/8, 0, 1/8, 2/8, 0, 0, 0, 0)$. The paper states this point is in $\text{conv}(P_6)$, gives a six-path witness, and identifies rigid F5 flows of $1/8$ and $2/8$. A separate min/max LP audit over the 60 enumerated vertices found exactly these two stage-5/6 pair flows fixed across all representations, leaving flexible mass $5/8$. Every x_6 coordinate is strictly below 1.

I independently enumerated the 60 valid $n=6$ pedigrees and verified an exact convex combination for this point. Its support, with weights in eighths, is $((13,12,23), 2)$, $((13,12,34), 2)$, $((13,34,12), 1)$, $((13,34,23), 1)$, $((23,34,13), 1)$, and $((23,34,24), 1)$, where each triple lists the common edges inserted at layers 4, 5, and 6. The standard-library check is in `verify_pedigree_example8.py`.

For the paper's constructed network, $z_{\text{Max}} = 5/8$ requires positive commodity flow, so the commodity list must be nonempty. But `MCFFeasible.src_val` requires $X(s.\text{arc_head}) = 1$ for every commodity, while every x_6 coordinate in this member is at most $3/8$. The Lean structure therefore cannot encode the paper's positive-flow MCF witness for this known member. This is a concrete completeness failure in the formalized input model, not a counterexample to the paper's mathematical theorem.

Shared capacity coupling is also absent from the Lean type

The paper's MCF includes shared constraints summing the flow of all commodities through each node and arc (equations 28 and 30). In `MCFFeasible`, `flow_le_slack`, `slack_nonneg`, and `col_sum_le` are each quantified separately for one commodity; the structure has no corresponding sum-over-commodities bound by `node_cap` and no aggregate-flow equality of the paper's kind. `flow_is_zMax` constrains only the total scalar flow. Thus the encoded feasibility predicate is not the paper's MCF even apart from the one-hot target restriction.

The Lean sufficiency theorem is only a special case

The comments in `N_Sufficiency.lean` outline assembling the witnesses for every commodity together with all rigid pedigrees. The actual theorem body obtains one commodity and returns the witness produced by `Y_s_in_conv`; it does not perform that assembly. With the one-hot condition above, the helper can certify a restricted single-destination situation. It does not formalize the stated general MCF sufficiency argument for fractional multi-destination inputs.

What would be needed to repair the claim

A valid verification would need to define `QuickProtocol` as an algorithm with a clear input/output specification, encode the paper's fractional MCF without forcing $X(s.\text{arc_head})=1$, and prove both directions of the membership characterization for the exact encoded network. It would also need to derive the runtime bound from that LP and show the separation/optimization reductions apply to the same polytope representation. Until those are supplied, the repository is useful as an evolving formalization scaffold, not a verified resolution of P versus NP.

Primary sources checked

Author's Lean repository README: <https://github.com/TiruArt/Pedigree-Polytopes-Lean4/blob/main/README.md>

N_PEqualsNP.lean: https://github.com/TiruArt/Pedigree-Polytopes-Lean4/blob/main/MembershipProject/Core/N_PEqualsNP.lean

N_LayeredNetworkTypes.lean: https://github.com/TiruArt/Pedigree-Polytopes-Lean4/blob/main/MembershipProject/Core/N_LayeredNetworkTypes.lean

N_YisinConv.lean and N_Sufficiency.lean: <https://github.com/TiruArt/Pedigree-Polytopes-Lean4/tree/main/MembershipProject/Core>

Arthanari (2025), Section 7.2: <https://arxiv.org/abs/2507.09069>

18. Late-September discrepancy update and a margin-normalized cover test

Current primary-source search found a 10 September 2026 arXiv preprint by Guo, Fang, and Lu, Vector Balancing via Directional Total Variation. Its Theorem 1.1 claims a dimension-free Komlos signing bound $K = 3 \sqrt{2 \pi}$. The authors say an automatic AI research agent discovered the proof. On 21 September, ETH researcher Afonso Bandeira posted a shorter exposition of the same result with constant 9π . These are very recent claims; this project has not independently audited the full proof. A further 20 September preprint claims a polynomial-time signing algorithm in a unit-cost exact-real-arithmetic model, which does not by itself establish polynomial bit complexity.

This discrepancy result is not a P-versus-NP result. The Clay Mathematics Institute continues to list P versus NP as unsolved. Sources: <https://arxiv.org/abs/2609.11189> ; <https://randomstrasse101.math.ethz.ch/posts/komlos/> ; <https://arxiv.org/abs/2609.23540> ; <https://www.claymath.org/problem/unsolved/>.

Proposition 18.1 - weighted-margin Komlos criterion

For each anchor a and coordinate i , let μ_a be a probability measure on a finite universe U , let $g_{(a,i)}(z) = +1$ when $z_i = a_i$ and -1 otherwise, and let $d_{(a,i)} = \sum_z \mu_a(z) g_{(a,i)}(z) > 0$. For every z , form the vector v_z with coordinates $\mu_a(z) g_{(a,i)}(z) / d_{(a,i)}$, and set $R = \max_z \|v_z\|_2$. Assuming a signing theorem with constant K , if $K R < 1$ then U has a partition E disjoint-union H such that both parts retain every anchor-coordinate strict majority. Therefore E and H belong to every existential majority-support filter in the family, while their empty intersection does not; a single pair covers the family.

Proof: choose signs ϵ_z from the balancing theorem applied to v_z / R . Then for every anchor-coordinate pair, $|\sum_z \epsilon_z \mu_a(z) g_{(a,i)}(z)| < K R d_{(a,i)} < d_{(a,i)}$. The signed margin on E and H is respectively half the total margin plus and minus this signed sum, so both are positive. This proves the required majority on both parts.

The criterion is heterogeneous in the margins: $\max_z \sum_{(a,i)} (\mu_a(z) / d_{(a,i)})^2 < 1 / (3 \sqrt{2 \pi})^2$ suffices under the new preprint's claimed constant. It sharpens the audit's earlier uniform-margin condition, but still misses tiny margins, many correlated heavy atoms, and general semi-filters with no majority-measure representation.

Finite sanity check: `experiment_weighted_margin.py` uses two disjoint radius-two Hamming balls around 0^{16} and 1^{16} . Each support has 137 points, the exact coordinate margin is $105/137$, and the normalized column radius is $4/105$. The claimed $3 \sqrt{2 \pi}$ criterion succeeds; a seeded signing was found on its first trial, and exact integer checks verified all 32 side-wise majorities. This is a toy set system, not an instance in the high-circuit-complexity promise domain. It does show a case missed by the older 36-constant test.

Research status: this extends a cover criterion and passes a finite consistency check. It establishes no asymptotic cover lower bound, no new P or NP separation, and no $P = NP$ algorithm. The remaining target is still a direct non-measure semi-filter packing or a new treatment of correlated heavy atoms. See Section 18 of `P_VS_NP_PROJECT_AUDIT_2026-08-24.md` for the full statement and analysis.

Proposition 18.2 - shared heavy core with diffuse tail

Let μ_a be a finite family of probability measures on U . If a common core S has mass $b_a = \mu_a(S)$ in $(0, 1/2]$ for every a , put $T = U$ minus S and $v_z = (\mu_a(z)/b_a)_a$ for z in T . Under a vector-balancing theorem with constant K , if K times $\max_z \|v_z\|_2$ is below 1, one signing of T has discrepancy $D_a = \sum_z \epsilon_z \mu_a(z)$ satisfying $|D_a| < b_a$ for every a .

Set $E = S$ union T_{plus} and $H = S$ union T_{minus} , where T_{plus} and T_{minus} are the two sign classes. Then $\mu_a(E) = b_a + (1 - b_a + D_a)/2 > 1/2$ and $\mu_a(H) = b_a + (1 - b_a - D_a)/2 > 1/2$. Their intersection is S , whose measure is $b_a \leq 1/2$. Thus this pair violates every strict-majority filter $F_a = \{W : \mu_a(W) > 1/2\}$. The core can contain several substantial atoms per measure.

A finite check is recorded in `experiment_heavy_core.py`. It uses three measures with five private atoms of mass $1/10$ each, for core mass exactly $1/2$, and a shared 100-point tail of mass $1/2$. The normalized radius is $\sqrt{3}/100$. The ECCC TR26-188 bound $K = 36$ applies: $36R = 9\sqrt{3}/25 < 18/25 < 1$, using $\sqrt{3} < 2$. A seeded signing was found on the first trial, and exact rational checks verified both cover-set measures exceed $1/2$ while the intersection has mass exactly $1/2$. The strict majority filter excludes that intersection. Source: <https://eccc.weizmann.ac.il/report/2026/188/>.

This extends the common-core case to its one-half boundary but does not cover families where no common core has mass in $(0, 1/2]$ for every measure, or where the remaining tail has large normalized atomic load. The experiment is synthetic and gives no Gap-MCSP lower bound or P-versus-NP result.

Corollary 18.3 - detecting the core by aggregate atom size

Let $w_z = (\mu_a(z))_a$ and, for a threshold $\theta > 0$, let $S_\theta = \{z : \|w_z\|_2 > \theta\}$. If some $b_0 > 0$ satisfies $b_0 \leq \mu_a(S_\theta) \leq 1/2$ for every measure, and $K\theta / b_0 < 1$, Proposition 18.2 applies: every tail atom has normalized vector norm at most θ / b_0 . With the ECCC constant $K = 36$, $\theta < b_0 / 36$ is sufficient. This reduces a family with several moderate or heavy atoms to a directly checkable common-core mass interval and a diffuse-tail bound.

For `experiment_heavy_core.py`, $\theta = 1/100$ separates private core atom norm $1/10$ from shared tail atom norm $\sqrt{3}/200$. Each measure assigns the core mass exactly $1/2$, so $b_0 = 1/2$ and $36\theta / b_0 = 18/25 < 1$. This covers multiple anchor-specific atoms at the half-mass boundary under the ECCC signing bound.

Scope remains limited: a common threshold core need not exist, the tail may have high normalized atom load, and arbitrary semi-filters may have no majority-measure representation. The result gives no asymptotic cover lower bound or P-versus-NP consequence.

Threshold-profile consequence: for $S_\theta = \{z : \|(\mu_a(z))_a\|_2 > \theta\}$, any family of majority measures evading every cover from Corollary 18.3 must, at every θ with $\max_a \mu_a(S_\theta) \leq 1/2$, have either $\min_a \mu_a(S_\theta) = 0$ or $\min_a \mu_a(S_\theta) \leq 36\theta$. Otherwise the ECCC $K = 36$ criterion supplies a cover. This only identifies the cases this sufficient method misses; it is not a general lower-bound criterion.

Example 18.4 - high-promise heavy-atom stress test

The heavy-core norm condition is sufficient, not necessary. Partition the $N = 2^n$ truth-table positions into three nearly equal blocks X, Y, Z and let x, y, z be their indicator tables. A counting argument shows that for any fixed $\beta < 1$, some such partition has all three tables of circuit complexity at least $s_2 = N^\beta$: the number of balanced partitions is $2^{(N \log_2 3 - O(\log N))}$, while bad partitions with one low-circuit table number at most $3 \cdot 2^{(2N/3 + o(N))}$. The anchor $a = 0^N$ is low complexity.

Put $\mu_1(x, y, z) = (49/100, 49/100, 2/100)$ and $\mu_2(x, y, z) = (2/100, 49/100, 49/100)$. At every coordinate exactly one table has bit one; under both measures, the mass agreeing with anchor bit zero is strictly greater than $1/2$. Thus both strict-majority filters are valid semi-filters above the same low anchor on the high-complexity promise.

The pair $E = \{x, z\}$, $H = \{y, z\}$ violates both filters: their E , H , and intersection masses are $(51/100, 51/100, 2/100)$ for μ_1 , and $(51/100, 49/50, 49/100)$ for μ_2 . Yet every common core with positive mass at most $1/2$ for both measures has normalized tail radius at least one, so the $K = 36$ common-core criterion fails. The aggregate threshold sets are U , $\{y\}$, or empty, and none meets Corollary 18.3.

`experiment_correlated_heavy_atoms.py` checks the exact rational masses, all candidate cores, and the explicit pair. The counting argument, not the script, establishes that x, y, z can be chosen in the high-complexity promise. This shows that large atom norms do not alone make a cover hard: correlated atoms can cancel under a single pair. A lower-bound construction in this regime must control those cross-filter cancellations.

Proposition 18.5 - small-support majority families have a short cover list

For M selected majority measures, each supported on at most t points with every atom below one half, a list of $L = \text{ceil}(3^t (\ln M + 1))$ pairs covers every filter. For each measure, partition its support into A, B, C so that $\mu(A)$ and $\mu(C)$ are below one half and $\mu(B)$ is at most one half. A uniform random 3-coloring matches that prescribed support labeling with probability at least 3^{-t} ; when it matches, $E = A \text{ union } B$ and $H = B \text{ union } C$ violate the filter. With L independent colorings, the probability that any of the M filters is missed is at most $M \exp(-L 3^{-t}) \leq e^{-1} < 1$, so such a list exists. If M is the number of low-complexity anchors, $M = 2^{O(N^\beta)}$, and $t = \alpha \log_2 N$, the list size is $N^{(\alpha \log_2 3 + \beta + o(1))}$. This rules out one selected small-support majority witness per anchor as a source of an $N^{(1+\epsilon)}$ cover lower bound whenever $\alpha \log_2 3 + \beta < 1 + \epsilon$. It does not bound arbitrary semi-filters or families with many measures per anchor.

Proposition 18.6 - aggregate-load condition for one three-part cover

For measures $\mu_1, \dots, \mu_M$ on U , set $w_z = (\mu_1(z), \dots, \mu_M(z))$ and $\delta = \max_z \|w_z\|_2$. Assume the Komlos signing guarantee: vectors of Euclidean norm at most one admit signs whose signed sum has infinity norm at most K . If $13 K \delta < 1$, a single partition $U = A \text{ disjoint-union } B \text{ disjoint-union } C$ has each measure of every part strictly below one half. Hence $E = A \text{ union } B$ and $H = B \text{ union } C$ satisfy $\mu_j(E), \mu_j(H) > 1/2$ and $\mu_j(E \cap H) < 1/2$ for all j . The ECCC TR26-188 constant $K=36$ gives the sufficient condition $\delta < 1/468$.

Proof: recursively apply the signing theorem to split a current block in two, so each child has measure at most half its parent's mass plus $K \delta/2$. After splitting U and then both halves, each of the four quarter-pieces has mass at most $1/4 + 3K \delta/4$. Split one quarter-piece once more: its two eighth-pieces each have mass at most $1/8 + 7K \delta/8$, while three quarter-pieces remain. Pair one eighth-piece with each of two quarter-pieces and use the last quarter-piece alone. Each paired part has mass at most $3/8 + 13K \delta/8 < 1/2$, and the unpaired quarter has mass at most $1/4 + 3K \delta/4 < 1/2$. Calling these parts A, B, C , the sets $E = A \cup B$ and $H = B \cup C$ each have measure above one half, while $E \cap H = B$ has measure below one half for every j . The numerical corollary depends on the signing result in the ECCC technical report; the deduction here is conditional on that guarantee.

Unlike the common-core criterion, this condition permits different measures to have largely disjoint diffuse supports. It fails for a shared heavy atom and becomes stricter when many measures reuse the same points. `experiment_small_aggregate_load.py` checks a finite family of 16 uniform measures on 4,096-point supports: the worst-case aggregate atom norm is at most $1/1,024$, below $1/468$, and one seeded 3-coloring was verified with exact integer counts. The experiment is not evidence of an asymptotic cover lower bound. For uniform supports of size t with each point in at most d selected supports, δ is at most $\sqrt{d}/t$, so the $K=36$ criterion gives one shared violating pair when $t > 468 \sqrt{d}$. More generally, maximum atom ρ and overlap d suffice when $468 \rho \sqrt{d} < 1$. The project has not established these low-overlap witness conditions for Gap-MCSP anchors.

An exact search in `experiment_split_tree_threshold.py` checks every full binary split-tree depth multiset with at most seven leaves and every grouping into three parts. It finds no improvement over $K \delta < 1/13$; for eight or more leaves, the sum of leaf upper bounds alone forces $K \delta < 1/(2(m-1)) \leq 1/14$. This is optimal only within repeated two-way signing followed by grouping leaves; direct three-way balancing remains open.

Proposition 18.7 - support-intersection coloring bounds selected majority-measure covers

Let $\mu_1, \dots, \mu_M$ be probability measures on U , each with finite support S_j and every atom strictly below $1/2$. Let G be their support-intersection graph, with edge ij exactly when S_i and S_j intersect. If $\chi(G)$ is its chromatic number, the selected majority filters $F_j = \{W \subseteq U : \mu_j(W) > 1/2\}$ have a cover list of at most $\chi(G)$ pairs.

Proof: Proposition 12.6 partitions each support $S_j = A_j \cup B_j \cup C_j$ disjoint-union with $\mu_j(A_j) < 1/2$, $\mu_j(C_j) < 1/2$, and $\mu_j(B_j) \leq 1/2$. Within a proper color class of G , supports are pairwise disjoint. Merge the local partitions into global A, B, C and assign points outside the supports arbitrarily. Each measure in the class keeps its local masses, so $E = A \cup B$ and $H = B \cup C$ both have mass above one half while $E \cap H = B$ has mass at most one half. One pair covers the color class; one pair per color covers all filters.

If support sizes are at most t and each point belongs to at most d selected supports, then the support graph has maximum degree at most $t(d-1)$, so the cover list has size at most $t(d-1)+1$. Pairwise disjoint supports need only one pair, without any Komlos threshold.

A finite check in `experiment_support_graph_cover.py` uses five uniform three-point measures whose support graph is a path plus an isolated vertex. A proper two-coloring gives two pairs that violate all five filters exactly; maximum point overlap is two.

Corollary 18.8 - high-sphere majority witnesses can be chosen disjointly

Fix $0 < \beta < 1$, $N = 2^n$, $s_2 = N^\beta$ and $s_1 = s_2/(cn)$ as in Proposition 12.4. Let Y be the low-circuit anchors and U the high-circuit promise set. For $k = \lceil K s_2 \rceil$ with K sufficiently large, every $V_a = \{z \in U : d_H(z, a) = k\}$ has size $2^{\Omega(s_2/n)}$, while $|Y| = 2^{O(s_2)}$; a uniform point of V_a disagrees with a at a fixed coordinate with probability $q_n = O(N^{-(\beta-1)})$. Choose odd $t \geq 3$ with $N 2^{t q_n ((t+1)/2)} = o(1)$, for example $t = \Theta(\log N)$ or a sufficiently large odd constant depending on β . Then there are pairwise disjoint S_a subset V_a , each of size t , with coordinatewise majority exactly a .

Proof: order Y and choose supports sequentially. Before processing a , fewer than $|Y|t = 2^{O(s_2)}t$ points have been used, an $o(1)$ fraction r_n of V_a . The remaining set has size $2^{\Omega(s_2/n)}$ and disagreement probability at most $q_n/(1-r_n)$. Draw t independent points from it. A union bound over coordinates makes the probability of any wrong majority at most $N 2^{t(q_n/(1-r_n))((t+1)/2)} = o(1)$; the probability of a repeat is at most $\text{binom}(t, 2)/2^{\Omega(s_2/n)} = o(1)$. Thus each anchor has a distinct good support avoiding all earlier supports.

The uniform measure on each S_a has every anchor literal above one half and atom size $1/t < 1/2$. Its support-intersection graph is edgeless, so Proposition 18.7 gives a one-pair cover. This rules out the natural one-sparse-witness-per-anchor repair using high-complexity spheres, including logarithmic supports large enough that Proposition 18.5's list exponent is not useful. It does not bound other measure choices, multiple measures per anchor, or arbitrary semi-filters. A toy check in `experiment_disjoint_sphere_witnesses.py` verifies the finite combinatorics; random flips there are not certified high-complexity tables.

19. Research frontier cross-check as of 23 September 2026

A primary-source scan adds neighboring results and two September advances in discrepancy theory. Proposition 18.7 bounds selected majority-measure covers by the chromatic number of their support-intersection graph, and Corollary 18.8 shows that one sparse high-sphere witness per anchor can be selected with disjoint supports and a one-pair cover. These results narrow the measure route but do not give the missing general Boolean circuit lower bound or promise fusion-cover lower bound.

Remote points: Ben Lee Volk, ECCC TR26-203 (22 September 2026), gives an optimal-remoteness deterministic polynomial-time algorithm over the rationals and an improved finite-field guarantee. This updates the project remote-point line, but the earlier full-span obstruction remains: low-circuit singleton truth tables span the entire truth-table space, leaving no remote table outside that span. Source:

<https://eccc.weizmann.ac.il/report/2026/203/>.

Range avoidance: Hanlin Ren and Ryan Williams, ECCC TR26-118 (July 2026), prove a near-maximum $2^n/n$ circuit lower bound for E^{prMA} with one bit of advice, using range avoidance, iterative win-win arguments, and PCP methods. The target class includes a promise-MA oracle and does not transfer here to NP or Gap-MCSP. Source:

<https://eccc.weizmann.ac.il/report/2026/118/>.

Monotone matching: Anup Rao, ECCC TR26-129 (revised 3 September 2026), proves $\exp(\Omega(\sqrt{n}))$ monotone circuit lower bounds for bipartite perfect matching. This is a strong restricted-model result; general algorithms may use nonmonotone operations, so it does not imply $P \neq NP$. Source: <https://eccc.weizmann.ac.il/report/2026/129/>.

Meta-complexity: Halley Goldberg, Mandar Juvekar, and Valentine Kabanets, ECCC TR26-091 (June 2026), prove conditional NP-hardness results for variants of implicit MCSP and full-support PAC learning under subexponential iO and proof-system assumptions, and rule out certain randomized Levin reductions. The result is conditional and does not establish unconditional standard-MCSP hardness. Source:

<https://eccc.weizmann.ac.il/report/2026/091/>.

Wire lower bounds: ECCC TR26-167 states near-cubic almost-everywhere wire lower bounds for two depth-two threshold circuit classes and an E^{NP} language. The report page records a substantive reader comment raising concerns about its exposition and relation to prior work. Its stated models do not give a general NP circuit lower bound, so the project draws no inference from it pending clarification. Source:

<https://eccc.weizmann.ac.il/report/2026/167/>.

Constructive Komlos signing: Guo, Fang, and Lu, arXiv:2609.23540 (20 September 2026), give a deterministic constant-discrepancy signing algorithm using polynomially many exact real-arithmetic operations, with time $O((m n^9 + n^{10}) \log(2+m+n))$. Substituting it into Proposition 18.6 makes the four recursive splits constructive for an explicitly supplied measure matrix. The model is exact real-RAM, and M and $|U|$ can be enormous in this project; this is not a polynomial-time SAT algorithm or a standard bit-complexity result.

Source: <https://arxiv.org/abs/2609.23540>.

Rank-two complex discrepancy: Guillen and Kobzar, arXiv:2609.15071v2 (revised 21 September 2026), prove a constant bound when signs may be arbitrary unit-complex phases. These phases are not restricted to three discrete colors, so the result does not directly supply the three-part strict-majority cover required here. Source: <https://arxiv.org/abs/2609.15071>.

Audit of a claimed topological collapse

M. Alasli arXiv:2603.22211 claims that $P=NP$ implies $\#P=FP$, but its proof has a direct gap. Lemma 3.1 says every function computed by a SAT algorithm is either computable from bounded-radius local information or distinguishes the cases $\beta_2=0$ and $\beta_2=2^{\Omega(N)}$. These alternatives are not exhaustive: being nonlocal does not imply distinguishing those two topological regimes. The proof therefore does not establish that a polynomial-time SAT decider computes a $\#P$ -hard topological function. Source: <https://arxiv.org/abs/2603.22211>.

The 2026 preprint relies on Alasli arXiv:2508.13200, which claims every 2-SAT solution cubical complex is contractible. Under its own definition, $F=(x \text{ OR } y) \text{ AND } (\text{NOT } x \text{ OR } \text{NOT } y)$ has exactly two satisfying vertices, 01 and 10, with no edge between them. The complex is disconnected, hence not contractible. This is an explicit counterexample to that background claim. Source: <https://arxiv.org/abs/2508.13200>.

These defects mean the topological-collapse preprint is not a proof of $P=NP$, $P \neq NP$, or $P=NP$ implying $\#P=FP$.

Current status: the Komlos constant-36 theorem used above is stated in Karingula and Lovett ECCC TR26-188 (17 September 2026). These external results enlarge the tool inventory but do not close the project central gap: a representation-independent family of semi-filters that defeats every short list of fusion pairs, or a different explicit lower-bound route with a complete transfer to $P \neq NP$. Source: <https://eccc.weizmann.ac.il/report/2026/188/>.

20. Re-audit of the proof-search deadline project notes

A local ChatGPT project mirror contains a finite self-reference route developed on 14 September 2026. Fix a sound arithmetic proof system T and a universal proof searcher U_T that accepts only verified T -proofs. The diagonal sentence D_n says U_T will not output a proof of D_n within 2^n steps. Under $\text{Con}(T)$, U_T misses every deadline, while a formalized finite nonhalting trace proves each D_n individually in T with a baseline proof-size upper bound $2^{O(n)}$.

The route proves a conditional implication: if these fixed sentences have sufficiently short T -proofs infinitely often, specifically $\liminf \log_2(1+s_n)/n = 0$ for minimum proof lengths s_n , then $\text{Con}(T)$ implies $P \neq NP$. Under $P=NP$, bounded-proof existence is decidable in polynomial time and a proof can be reconstructed in time polynomial in its length; universal simulation then contradicts the missed deadlines. The required short-proof condition remains unproved, and the $2^{O(n)}$ baseline is too large.

The paired refutation note does not prove the short-proof condition false. It shows that an approach passing through a full bounded-consistency intermediate statement inherits a large proof-length lower bound, but no size-controlled reduction from arbitrary proofs of D_n to that intermediate statement is known. Its oracle comparison builds a different proof system with an oracle and added axiom scheme, where short proofs coexist with missed deadlines. This rules out a generic inference from soundness and universal search alone; it does not refute the original PA-family condition.

This is a conditional proof-complexity project, not a P-versus-NP breakthrough. It supplies explicit true, provable sentences, but the subexponential proof-length upper bound needed for a separation has not been established. Sources in the project notes include Krajicek on algorithmic proof search, <https://arxiv.org/abs/2104.04711>; Pudlak on finite consistency proof lengths, <https://users.math.cas.cz/~pudlak/fin-con.pdf>; and Barak on proof search if $P=NP$, https://introcs.org/public/lec_14_PvsNP.html.

The local project mirrors available here do not contain a complete export of all private ChatGPT project conversations. This section audits the proof-deadline notes present in the mirror and does not claim to cover unexported chats.

21. Exact preservation closure and new finite stress tests

The cylinder-generated filter candidate was exhaustively checked and rejected. The same work gives an exact closure formulation for testing whether a proposed list of pairs covers every semi-filter above a collection of required generators. This formulation applies beyond majority-measure filters.

Proposition 21.1 - preservation closure characterizes all-filter covers

Let U be finite. For an anchor a , let $G(a,1), \dots, G(a,r)$ be the required nonempty generators, and let Λ be a finite list of pairs (E,H) of subsets of U . Set $C(a,0)$ to the upward closure of the generators. Recursively set $C(a,m+1)$ to the upward closure of $C(a,m)$ together with every intersection $E \cap H$ for which (E,H) is in Λ and both E and H are already in $C(a,m)$. The process stabilizes because U is finite. Λ covers every semi-filter above a exactly when the stabilized family contains the empty set.

Proof. Every semi-filter above a contains the generators and is upward closed. If it preserves every pair in Λ , induction shows it contains every set added by the closure. Thus a

closure containing the empty set leaves no preserving semi-filter. If the empty set is not reached, the stabilized family itself is upward closed, contains U , excludes the empty set, contains the generators, and preserves every pair in Λ . It is therefore a semi-filter preserving the entire list.

This gives an explicit counterexample to any proposed list that fails: the stabilized closure is itself a semi-filter preserving that list. Equivalently, the cover problem asks for a small set of intersection rules that forces the forbidden empty set from the anchor generators.

experiment_toy_all_semfilters.py verifies the equivalence in all 1,024 pair-anchor cases of a four-point example.

Corollary 21.2 - exact criterion for a one-pair cover

A single pair (E,H) covers all semi-filters above all anchors exactly when E intersect H is empty and, for every anchor a , both E and H contain at least one required generator for a . Necessity follows because the maximal semi-filter consisting of all nonempty subsets preserves every pair with nonempty intersection, while a filter above a is forced to contain a set W exactly when W contains one of its required generators. The same facts prove sufficiency.

For the Gap-MCSP bit-literal generators, fixing at most two of the $N=2^n$ truth-table coordinates leaves at least $2^{(N-2)}$ completions. The number of tables with circuit complexity below $s_2=N^\beta$ is at most $2^{(O(s_2 \log(n+s_2)))}=2^{o(N)}$. Hence every intersection of two required literal slices contains a high-complexity table. No one pair can therefore cover the entire arbitrary semi-filter family: any pair forced into every filter above an anchor has nonempty intersection and is preserved by the maximal semi-filter. This proves only $\rho \geq 2$, not a magnification-scale lower bound.

Exact four-point instance: arbitrary semi-filters require two pairs

Let U be the four three-bit strings of weight at least two; anchors are the four strings of weight at most one. Exhaustively enumerating all 166 semi-filters on U and retaining those above each anchor gives 17 anchor-filter instances. There are 256 ordered pairs (E,H) ; the best one violates 16 filters, and the exact minimum cover is two:

$(E_1,H_1)=(\{011,110\},\{101,111\})$ and $(E_2,H_2)=(\{101,110\},\{011,111\})$. Each pair hits 16 filters, and together they hit all 17.

The first pair misses the filter above anchor 010 generated upward from the singleton sets $\{011\}$ and $\{110\}$. That filter contains E_1 but not H_1 , so it preserves the pair; the second pair violates it. This finite example shows that arbitrary semi-filters can require more pairs than the canonical subfamily, but its universe is too small to imply an asymptotic bound. See experiment_toy_all_semfilters.py.

Cylinder-filter stress test: rejected

For U the eleven four-bit strings of weight at least two and the five anchors of weight at most one, define $F(a,k)$ as the upward closure of every nonempty cylinder fixing between one and k coordinates to anchor values. Exhaustive enumeration over all E,H subsets of U finds a one-pair cover for each $k=1,2,3,4$. For k at least two, a disjoint pair is $E=\{0110,1001\}$, $H=\{0101,1010\}$. This rejects the cylinder-generated variant; it does not address arbitrary non-measure semi-filters. See experiment_cylinder_filter_cover.py.

Five-point controls: both cover behaviors occur

On $U = \{z \text{ in } \{0,1\}^4 : \text{weight}(z) \text{ at least } 3\}$ with anchors of weight at most 2, all 7,579 semi-filters on the five-point U yield 136 anchor-filter instances. Exact enumeration finds minimum cover two; the one-pair maximum hits 129 filters. The two disjoint pairs are $E1 = \{0111, 1011\}$, $H1 = \{1101, 1110\}$ and $E2 = \{0111, 1101\}$, $H2 = \{1011, 1110\}$. Together they cover all filters. See `experiment_toy_all_semfilters_5point.py`.

As a positive geometry control, take U to be the five 5-bit strings of weight exactly four and anchors of weight at most one. The same 7,579 valid semi-filters produce 11 anchored instances, all covered by one disjoint pair: $E = \{01111, 10111\}$, $H = \{11011, 11101\}$. Here both sides contain a required generator for every anchor. See `experiment_toy_shell_semfilters.py`. This tiny model does not lift to the dense high-circuit promise, where every two-coordinate cylinder remains nonempty.

Proposition 21.3 - sparse co-singleton shells have a one-pair cover

For $n \geq 2r+2$, let U_n be the n Boolean strings 1^{n-e_i} with exactly one zero, and let $Y_{(n,r)}$ be all anchors of Hamming weight at most r . Partition $[n]$ into I and J with each side of size at least $r+1$. Set $E = \{1^{n-e_i} : i \text{ in } I\}$ and $H = \{1^{n-e_i} : i \text{ in } J\}$. Then (E,H) violates every semi-filter above every anchor in $Y_{(n,r)}$.

For coordinate i , the anchor-matching literal slice in U_n is the singleton $\{1^{n-e_i}\}$ when $a_i=0$, and U_n minus that singleton when $a_i=1$. Each part has more than r points, so it contains a point whose zero coordinate is outside the support of a . Each side therefore contains a required generator for every anchor. The two sides are disjoint, so their intersection is empty and every semi-filter excludes it. `experiment_toy_shell_semfilters.py` verifies the generator containments for 23 parameter pairs with $n=4..10$ and exhaustively checks the smallest $n=5$, $r=1$ case.

This proves a one-pair cover for a sparse toy promise and identifies its key geometric condition. The singleton-slice property fails for the real Gap-MCSP universe: every one- or two-coordinate slice contains exponentially many high-complexity tables. No transfer to the target cover lower bound follows.

Proposition 21.4 - global anchor-support form of preservation closure

Let A be a finite anchor set, let each anchor a have required nonempty generators G_a in a finite universe U , and let Λ be a list of pairs (E,H) . Define A_S^t as the anchors whose preservation closure contains S at stage t . Then $A_S^0 = \{a : \text{some } G_a \text{ is a subset of } S\}$, and $A_S^{t+1} = A_S^t \cup \bigcup_{(E,H) \text{ in } \Lambda} (E \cap H \text{ is a subset of } S \text{ of } (A_E^t \cap A_H^t))$. For every stage and set S , this gives exactly the same membership as running the preservation closure separately for each anchor. It is enough to store S in $\{\text{empty set}\} \cup \{E, H, E \cap H : (E,H) \text{ in } \Lambda\}$, with at most $3|\Lambda|+1$ sets. Thus Λ covers every semi-filter above every anchor in A exactly when A is contained in A_{empty}^∞ .

The proof is induction on closure stages: S is already present, or a listed rule has both endpoints present and its intersection is contained in S . Passing to anchor sets turns the endpoint condition into $A_E \cap A_H$. This is an exact joint checker that avoids enumerating all $2^{|U|}$ subsets or semi-filters. The experiment

experiment_anchor_support_closure.py compares this recurrence with direct closure for every one- and two-rule list in the four-point model (32,896 lists), and for 2,048 seeded multi-rule lists plus all single rules in the five-point model (3,072 total); all tested memberships agree. The anchor-support sets may remain exponentially large for the low-circuit anchor family, so this sharpens the representation of the obstruction without proving a lower bound.

Fractional minimax stress test - no scalable witness found

On the four-point and five-point threshold toys, a numerical linear program searched for an anchor-balanced distribution on anchored semi-filters minimizing the largest mass captured by any ordered pair. The solver returned $3/4$ and $9/11$, respectively. In each case its selected support has one equally weighted filter per anchor; exhaustive enumeration of all ordered pairs gives exact maxima of 3 of 4 and 9 of 11 filters on those supports. The LP values have no separately supplied exact dual certificate, and these finite examples imply no asymptotic trend. They do not give the vanishing single-pair influence needed for a fractional-cover proof at magnification scale. See experiment_toy_semfilter_minimax.py.

Transfer test for recent local decision trees - only the baseline depth

Golovnev and Gurumukhani's ECCC TR26-195 local-decision-tree model looked relevant because sufficiently strong adaptive lower bounds imply circuit lower bounds. For Gap-MCSP, a direct counting argument gives only the baseline. Fix an accepting low-circuit table y on a correct depth- d ℓ -local decision tree. At most $d\ell$ input positions are queried along its path, and every completion agreeing with y on those positions follows the same path. There are at most $2^{O(s^2 \log(n+s^2))}$ truth tables with circuits below s^2 , so the completion subcube must be no larger than that count; otherwise it contains a high-circuit table, contradicting correctness. The path therefore touches at least $N - O(s^2 \log(n+s^2))$ positions, giving $d\ell \geq N - O(s^2 \log(n+s^2))$. With $s^2 = N^\beta$ for $\beta < 1$, this is $(1 - o(1))N/\ell$, essentially the trivial N/ℓ bound. It gives no superconstant depth overhead and does not activate the paper's circuit-lower-bound implication. See <https://eccc.weizmann.ac.il/report/2026/195/>.

Proposition 21.5 - firing-rule circuits compile covers into promise separators

Let $\Lambda = \{(E_j, H_j) : j \in [m]\}$, $K_j = E_j \cap H_j$, and $b_X(a)$ indicate that at least one required generator for anchor a is contained in X . Set $F_j^0(a) = 0$ and define $A_X^t(a) = b_X(a) \text{ OR the OR of } F_j^t(a) \text{ over } j \text{ with } K_j \subset X$, for every endpoint X and $X = \emptyset$. Then $F_j^{t+1}(a) = F_j^t(a) \text{ OR } (A_{E_j^t}(a) \text{ AND } A_{H_j^t}(a))$. $A_{\emptyset}^m(a) = 1$ exactly when closure derives empty: each fired rule adds only its fixed intersection, and at most m rule flags can newly fire, so m rounds suffice.

For bit-literal generators, $b_X(a)$ is an OR of at most $2N$ input literals, with $a_i = b$ included when the high-table slice $B_{i,b}$ is contained in X . Hardwiring these flags costs $O(Nm)$ gates over at most $2m$ endpoint sets. Unrolling m rounds costs $O(m^3)$ additional gates. Thus a size- m universal cover yields a nonuniform De Morgan circuit separator of size $O(Nm + m^3)$: it accepts low-circuit anchors, while every high-circuit anchor z has the principal semi-filter

$\{S \subset U: z \in S\}$, which preserves every pair. The firing recurrence matched direct closure on 32,896 four-point and 3,072 five-point rule-list instances in `experiment_anchor_support_circuit.py`. The cubic overhead and hardwired slice containments do not yield the required magnification-scale lower bound.

Proposition 21.6 - exact two-pair cover at the sparse-shell boundary

Let $n=2r+1$ with $r \geq 1$, let $u_i = 1^n - e_i$ be the n strings with exactly one zero, add $v = 1^n$, and take anchors of weight at most r . For anchor a , its matching coordinate generator is $\{u_i\}$ if $a_i=0$, and U minus $\{u_i\}$ if $a_i=1$. The minimum universal cover list over all semi-filters has exactly two pairs.

For the upper bound, partition coordinates into I, J with $|I|=r+1$ and $|J|=r$. The disjoint pair $E=\{u_i: i \in I\}$, $H=\{u_j: j \in J\}$ covers every anchor except the one whose zero set is exactly I : every anchor has at least $r+1$ zeros, so its zero set intersects I , and it intersects J unless it equals I . A second disjoint pair $(\{u_p\}, \{u_q\})$ for any distinct p, q in I covers that exceptional anchor.

For the lower bound, any universal single pair must be disjoint and each side must contain a matching generator for every anchor. For a fixed anchor, two matching generators can be disjoint only when they are zero-slices $\{u_i\}, \{u_j\}$ at distinct zero coordinates: two one-slices intersect at v , and a one-slice intersects every different zero-slice. The singleton indices in each side must therefore hit every $(r+1)$ -subset of $[n]$, requiring at least $r+1$ indices per side. Since the sides are disjoint this needs $2r+2$ coordinates, but $n=2r+1$. Thus one pair cannot suffice.

The six-point experiment `experiment_toy_all_semfilters_6point.py` ($n=5, r=2$) exhaustively finds that one pair covers at most 15 of 16 anchors and verifies the explicit two-pair cover. This theorem concerns a sparse promise with singleton zero-slices. It does not transfer to Gap-MCSP, where every two-coordinate literal cylinder contains exponentially many high-circuit tables.

Proposition 21.7 - proof-DAG leaves give a near-linear Gap-MCSP cover lower bound

Let Λ contain m pair rules, and fix a low-circuit anchor a . If its preservation closure derives the empty set, choose the first rule application q whose fixed intersection $K_q = E_q$ intersection H_q is empty. Trace each endpoint's first derivation. An endpoint is witnessed either by one initial matching literal generator contained in it, or by a rule j that fired earlier and whose fixed intersection K_j is contained in it. Memoizing each earlier rule gives an acyclic AND-DAG with at most m rule nodes, each with two antecedents. The DAG is connected to q , so it has at most $m+1$ generator leaves: with k rule nodes and L leaves, it has at most $2k$ edges and connectedness requires at least $k+L-1$, so $L \leq k+1 \leq m+1$. Induction up the DAG shows that the intersection of all leaf generators is contained in K_q , hence is empty in U .

Every leaf is a matching bit-literal generator for the same anchor, so the leaves fix a consistent set Q of at most $m+1$ truth-table coordinates. Their intersection is exactly the high-circuit promise restricted to the subcube that agrees with a on Q . Let M_2 be the number of truth tables computed by circuits of size below s_2 . If $2^{N-|Q|} > M_2$, that subcube contains a high-circuit table, contradicting the empty intersection. Therefore $\rho_{\text{GapMCSP}}(n, \beta) \geq$

$N - \log_2(M_2) - 1 = N - O(s_2 \log(n + s_2)) - 1$, using the standard circuit-counting bound $M_2 \leq 2^{O(s_2 \log(n + s_2))}$.

For $s_2 = N^\beta$ with fixed $\beta < 1$, the error is $o(N)$, so this gives $\rho \geq (1 - o(1))N$ for the all-semi-filter cover measure. This is the first size-dependent lower bound in the audit, but it remains linear and falls short of the $N^{(1+\epsilon)}$ magnification threshold. The proof-DAG leaf argument cannot exceed N because there are only N truth-table coordinates; a stronger result must exploit structure beyond the number of literal leaves. The experiment `experiment_proof_dag_leaf_bound.py` traces all 263,168 anchor/rule-list cases for all one- and two-rule lists in the four-point toy; all 45,606 empty derivations have empty leaf intersection and at most $k+1$ leaves for k proof rules. This finite check validates the trace implementation, not the asymptotic theorem. This does not prove $P \neq NP$.

Proposition 22.1 - projected affine controls give a 2^r poly(N) SAT algorithm

Represent a circuit with XOR, NOT, and binary AND gates using one variable per wire. Put the XOR/NOT equations and $\text{output}=1$ into an affine $\text{GF}(2)$ system A , leaving out the AND equations. Choose one input wire from each AND gate; let B be the set of distinct chosen wires and $r = \dim \pi_B(\text{Sol}(A))$. If A is inconsistent, reject. Otherwise enumerate the 2^r patterns in this affine projection and fix the chosen wires. Each AND equation $z = x \text{ AND } y$ becomes affine: $z=0$ when its chosen input is 0, and z equals the other input when the chosen input is 1. Gaussian elimination decides each resulting system, giving a $2^r \text{ poly}(N)$ -time algorithm. Since $r \leq |B| \leq k$ for k AND gates, this also gives the standard $2^k \text{ poly}(N)$ bound. For a supplied control choice, r is computable by linear algebra. If the minimum projected rank is $O(\log N)$, the combined solver is polynomial time. Section 22.3 gives an FPT method to find optimal controls, but it assumes native XOR/AND structure and does not recover affine constraints from arbitrary CNF.

Proposition 22.2 - minimum projected control rank is NP-hard to find

Given a simple graph $G=(V,E)$ with at least one edge, create input bits x_v and an AND gate $y_{uv} = x_u \text{ AND } x_v$ for every edge, then XOR all y_{uv} and require $\text{output}=1$. The affine relaxation A constrains only the AND-output variables through the XOR accumulator and output requirement; it leaves all primary inputs free. Therefore every chosen control set B of primary input wires has projection $\pi_B(\text{Sol}(A))$ equal to the full Boolean cube and rank $|B|$. Choosing one input of each AND gate selects one endpoint of every edge, so the selected wire set is a vertex cover. Conversely, any vertex cover can select one endpoint for each edge. Thus the minimum projected rank is exactly the minimum vertex-cover size. This is an acyclic circuit with every AND gate in the output cone, so exact optimization of r is NP-hard in classical complexity. A matching of k disjoint edges gives $r^*=k$ for an $O(k)$ -gate circuit, so there is no universal logarithmic rank bound. This does not preclude fixed-parameter tractability in r ; the next result identifies the general problem and gives the corresponding branch search.

The independent experiment `experiment_branch_rank_vertex_cover.py` enumerates all 1,098 simple graphs on 2-5 vertices and 59,808 control selections; the minimum rank always equals brute-force vertex-cover size. The general result follows from the reduction, not the finite check.

Proposition 22.3 - control selection is Rank Vertex Cover and is FPT in r

First replace each AND gate with identical inputs, $z=x \text{ AND } x$, by its affine equation $z=x$ and include it in A . Assume the resulting affine system is consistent, with solution set x_0+L and $L=\ker(A)$. Choose a basis $v_1,\dots,v_d$ for L and associate to each wire w the vector $c_w=((v_1)_w,\dots,(v_d)_w)$. For every wire set B , $\dim \pi_B(\text{Sol}(A))$ equals the linear rank of $\{c_w:w \text{ in } B\}$. Make a graph with circuit wires as vertices and one edge between the distinct input wires of every remaining AND gate; merge duplicate edges. A valid control set is exactly a vertex cover, and its rank is exactly the projected affine rank.

At each search node, maintain a basis for the span of selected control vectors. If some AND edge has both endpoint vectors outside the span, branch on either endpoint; each branch raises rank by one. If no such edge remains, every AND has an endpoint in the span, and those endpoints form a cover of rank at most the current budget. For completeness, a rank- q cover always contains an endpoint of every uncovered edge, so following that endpoint keeps the current span inside the cover's span and reaches a successful node within q branches.

With rank budget q , this search has depth at most q and at most $2^{(q+1)}-1$ nodes. Iterative deepening finds a minimum-rank control cover r^* in $2^{O(r^*)} \text{poly}(N)$ time. Enumerating its 2^{r^*} affine projection patterns and solving each slice by Gaussian elimination takes the same asymptotic time. Thus full control discovery plus SAT decision is $2^{O(r^*)} \text{poly}(N)$, with an exponential worst case when r^* is large.

The rank-cover check in `experiment_affine_control_rank_fpt.py` matched exhaustive minima on all 1,098 identity-matroid graphs through five vertices, all 16,384 simple four-vertex graphs paired with binary vector representations, and 1,000 random represented-matroid instances. The integrated affine-control SAT solver in `experiment_affine_control_rank_sat.py` matched exhaustive SAT on 2,500 random small systems, with 1,039 SAT and 1,461 UNSAT. It also solved shared-control instances with 8, 64, and 256 AND gates at rank one, using three search nodes each. These checks validate the implementation only.

Rank Vertex Cover is established parameterized-complexity work. Meesum, Panolan, Saurabh, and Zehavi use it for algebraic compression (<https://arxiv.org/abs/1705.02822>); a 2024 ESA paper notes that Rank Vertex Cover is folklore FPT parameterized by rank (<https://drops.dagstuhl.de/storage/00lipics/lipics-vol308-esa2024/LIPIcs.ESA.2024.17/LIPIcs.ESA.2024.17.pdf>). The project contribution here is recognizing that the affine projection rank is exactly this known objective and composing control search with affine-slice SAT; the FPT branching principle itself is not claimed as new.

The remaining issue is the value of r^* on unrestricted circuits and whether useful affine structure can be recovered from arbitrary SAT encodings. No universal small-rank theorem, polynomial-time all-SAT algorithm, or P-versus-NP consequence emerged. This is an exact parameterized algorithm for a structured slice of SAT, not a breakthrough resolving the open problem.

Ben Daniel's ECCC TR26-201 studies witness isolation from circuit descriptions. Under indistinguishability obfuscation, a randomized polynomial-size pruning procedure that isolates every nonempty affine-subspace input with probability at least $a/\log L$ would imply $\text{NP} \subseteq \text{P}/\text{poly}$; iO together with nonuniformly secure one-way functions rules out such an isolator. In a membership-query-only model, subexponential query budgets still allow only $\Theta(1/n)$ worst-case isolation probability on affine targets. This is a conditional barrier to generic witness-picking routines, not an unconditional P-versus-NP result, and it does not

apply to the exact $2^{O(r^*)} \text{poly}(N)$ decision algorithm above. See <https://eccc.weizmann.ac.il/report/2026/201/>.

Proposition 23.1 - Reed-Muller feature-span filters have a one-pair cover

Let $\Omega = \{0,1\}^N$, let $A = \Omega$ minus U be all tables below the high-complexity threshold s_2 , with $|A| \leq M = 2^{(O(s_2 \log(n+s_2)))} = 2^{o(N)}$ for $s_2 = N^\beta$ and fixed $\beta < 1$. The low-circuit anchors Y are a subset of A . Let $d < N$ and map each x to $\phi_d(x)$, the vector of all squarefree monomials of degree at most d evaluated at x ; its dimension is $D = \sum_{j \leq d} \binom{N}{j}$. The proof applies when $M < 2^{(N-1-d)}$ and $D+1 < 2^{(N-d)} \cdot M$. These inequalities hold for every $d = o(N)$; for fixed d/N tending to $\alpha < 1/2$, the second holds when $H_2(\alpha) < 1 - \alpha$, where H_2 is binary entropy. The equality occurs at $\alpha^* \approx 0.22709$. For each low-circuit anchor a in Y define $F_{\{a,d\}} = \{S \subset U : \phi_d(a) \text{ lies in the affine } F_2\text{-span of } \phi_d(S)\}$, taking the affine hull of the empty set to be empty. These families are upward closed and exclude the empty set, so they are semi-filters. Every required literal generator $G_{\{i,a_i\}} = U \cap \{x : x_i = a_i\}$ belongs to $F_{\{a,d\}}$. If not, affine separation gives a nonzero degree- d Boolean polynomial that is 1 at a and vanishes on all high-circuit points in that coordinate slice. Its restriction is a nonzero Reed-Muller codeword $RM(d, N-1)$, with weight at least $2^{(N-1-d)}$, but its support would lie among at most M low tables, contradiction. Likewise $\phi_d(U)$ spans $\phi_d(\Omega)$, since a nonzero degree- d polynomial has weight at least $2^{(N-d)} > M$. The Reed-Muller minimum-distance formula is classical; see Delsarte, Goethals, and MacWilliams (1970), [https://doi.org/10.1016/S0019-9958\(70\)90214-7](https://doi.org/10.1016/S0019-9958(70)90214-7).

The feature image has affine dimension $r = D-1$, with at most $2^{(r+1)} = 2^D$ affine hyperplanes. Each proper hyperplane misses at least $\delta = 2^{(N-d)} \cdot M$ points of U . Color U independently with two colors. For one specified color and one hyperplane, the probability that all outside points receive that color is at most $2^{(-\delta)}$; a union bound over both colors and all hyperplanes gives failure probability at most $2^{(D+1-\delta)} < 1$, since $D = 2^{o(N)}$ and $\delta = 2^{(N-o(N))}$. Thus there is a disjoint partition $U = E \cup H$ with both parts spanning the full feature space. Every low anchor feature lies in both spans, so E, H belong to every $F_{\{a,d\}}$ for a in Y , while $E \cap H$ is empty and is excluded. One pair covers the entire low-degree feature-span family.

The affine experiment `experiment_affine_span_filter_failure.py` checks all 696 excluded sets of sizes one through three in a four-bit cube. The Reed-Muller experiment `experiment_low_degree_span_filter.py` uses degree-two features on an eight-bit cube (37 monomials), checking all 256 single-anchor exclusions and 512 seeded exclusions of sizes two or three. All literal-slice membership and full-span partition checks passed. The finite checks validate the code only; the asymptotic conclusion follows from Reed-Muller distance and the probabilistic union bound.

This rules out affine and Reed-Muller feature-span filters satisfying the distance-versus-dimension conditions as sources of a cover lower bound. In particular it covers every $d = o(N)$ and every fixed degree fraction below $\alpha^* \approx 0.22709$. This entropy cutoff is where this union-bound proof stops, not a demonstrated boundary for the filter family. It does not upper-bound the full cover complexity or prove $P \neq NP$: arbitrary semi-filters need not arise from feature-space span, and an uncovered filter against every short pair list remains to be constructed.

Relevant literature updates, 4-23 September 2026

Haoxing Lin, ECCC TR26-139, revision 2 (4 September 2026), proves unconditional fine-grained AC0 size lower bounds for k-OV, k-XOR, and k-SUM through depth-zero projections from colored subgraph isomorphism. At fixed depth the bounds scale as $n^{\Omega(k)}$ for the stated parameter ranges, with additional growing-k bounds for selected gate orientations. These results concern bounded-depth circuits and fine-grained targets; they supply no transfer to general Boolean circuits or Gap-MCSP, so they do not provide the superlinear cover lower bound needed by the magnification route. See <https://eccc.weizmann.ac.il/report/2026/139/>.

Golovnev and Gurumukhani, ECCC TR26-195, Sumset Structure in Local Computation (<https://eccc.weizmann.ac.il/report/2026/195/>), introduce local decision trees. Strong lower bounds for their general adaptive model would imply significant circuit lower bounds; their explicit near-optimal results concern an oblivious restriction. This is a promising intermediate-model direction, but it gives no SAT lower bound.

Li, Ren, and Zhong, ECCC TR26-196, Many Proof Complexity Generators Inside One Demi-Bits Generator (<https://eccc.weizmann.ac.il/report/2026/196/>), show that random output projections can turn demi-bits generators into proof-complexity generators and derive restricted range-avoidance consequences, including an LPN-conditional result. It does not provide the needed generator or an unconditional separation.

Kayal et al., ECCC TR26-206, Certification Complexity of Boolean Functions (<https://eccc.weizmann.ac.il/report/2026/206/>), develop operational certification measures and new exact and zero-error quantum query lower bounds. These results concern query complexity, not general Boolean circuit lower bounds.

Carmosino, Gupta, and Volkovich, ECCC TR26-207, Towards an Interesting VPSPACE-complete Problem (<https://eccc.weizmann.ac.il/report/2026/207/>), study algebraic polynomial families connected to TQBF and the permanent. Small arithmetic circuits for the families imply strong class containments or a PSPACE subset NP with a PIT oracle; these implications do not follow from $P=NP$ alone.

The Clay Mathematics Institute still lists P vs NP as unsolved (<https://www.claymath.org/millennium/p-vs-np/>). No positive P-versus-NP result has been obtained here. The strongest new outputs are exact closure characterizations, a firing-rule circuit compilation, a sparse-shell boundary theorem, and a near-linear all-semi-filter cover lower bound for Gap-MCSP; the superlinear lower bound needed by the magnification route remains open. The affine-plus-AND line yields an exact SAT algorithm in $2^{O(r^*)} \text{poly}(N)$, where r^* is the minimum projected affine control rank. Finding optimal controls is classically NP-hard, but it is exactly Rank Vertex Cover and is FPT in r^* , so controls can be found within the same parameterized bound. Exhaustive and randomized checks passed; no small-rank bound holds for all circuits, and this is not a P-versus-NP proof. The Reed-Muller feature-span result also rules out a broad algebraic candidate family with a one-pair cover.